



INFO

 27/02/2002
 Porto, Portugal
 +351 961 903 583
 kikopiressilva@gmail.com
 xizesse.github.io
 /in/franciscopiressilva
 /Xizesse

AREAS

Mobile robotics · Marine robotics · Motion planning · Decision making · Obstacle detection and tracking · Control system design

TOOLS

ROS · ROS 2 · Nav2 · Gazebo · ArduPilot · ArduSub · LiDAR · Codesys · MATLAB · CAD · Linux · Git · LaTeX

LANGUAGES

C++ · C · Python · Java

SPOKEN

Portuguese (native) · English (fluent) · German (beginner)

Francisco Pires da Silva

Robotics Engineer — Mobile & Marine Robotics

I'm a robotics engineer and researcher with an MSc in Electrical and Computer Engineering, currently working on autonomous surface vehicle navigation: obstacle detection, motion planning and avoidance. Drawn to mobile robotics by the complexity of the problems, the creativity the solutions demand, and the range of fields it can transform.

Experience

Robotics Researcher

INESC TEC — Centre for Robotics and Autonomous Systems (CRAS) 

Jan 2023 – Present | Porto, Portugal

- Developed and validated the full autonomous navigation stack for an ASV — motion planning, decision making and control system design.
- Dynamic obstacle avoidance for heterogeneous surface and aerial vehicles, with LiDAR-based obstacle detection and tracking.
- Collaborative UAV-ASV mission coordination for offshore inspection.
- 30+ days of field testing, deployments and live demonstrations on the water.

Horizon Europe / H2020: AEROSUB (offshore wind inspection) · TALOS (photovoltaic inspection) · ATLANTIS (Atlantic maritime robotics test platform).

Education

PhD Coursework, Electrical and Computer Engineering

Sept 2025 – Aug 2026 | University of Porto (FEUP)
Curricular year completed.

Exchange Semester, Electrical and Computer Engineering

Oct 2024 – Mar 2025 | Technical University of Munich (TUM), Germany

MSc in Electrical and Computer Engineering

Sept 2023 – Jul 2025 | University of Porto (FEUP) | GPA 18/20
Dissertation: *A Velocity Obstacle Avoidance Algorithm for Heterogeneous Robotic Vehicles* — 19/20.

BSc in Electrical and Computer Engineering

Sept 2020 – Jul 2023 | University of Porto (FEUP) | GPA 18/20

MIT Portugal Marine Robotics Summer School

Jul 2026 | Faial, Azores

Leadership & Outreach

NEEEC-FEUP

Sept 2020 – Aug 2025 | Electrical and Computer Engineering Students' Branch

- Board Member 2023/24 — headed the technology department, leading a team of 20.
- Logistics coordinator for NEEEL-IT, a robotics competition.
- Ran technical workshops (ROS 2, electronics, Arduino) and high-school outreach.

A little more about me, and the projects I've been involved in:

 xizesse.github.io